

NASA BRINGS REAL WORLD PROBLEMS FOR INDIAN YOUNGSTERS TO SOLVE

by Prasad Banerjee

You know how they say NASA is filled with Indians? Well, they may have a few more to come. The International Space Apps Challenge by NASA was held in Delhi this month. The event invites developers, many in their early 20s, to come and design apps for NASA. It's what we all know as a hackathon, and NASA arranged for a bunch of challenges, including Robotics, Outer Space, Humans and more. The projects would be judged at a global level by NASA, based on Best Use of Hardware, Best Use of Data, Most Inspiring, Galactic Impact, Best Mission Concept. Two winners would be chosen from each chapter, with an additional People's Choice Award as well.

Let's not dwell on the rules though, there's more to this than that. The winners of the Delhi chapter would go on to compete at the global level, against chosen candidates from all over the world.

There were two winners in the Delhi chapter. The judges chose a project named Spock and another named Glove Above. In fact, Glove Above also took away the People's Choice Award with them. Both space gloves, they deal with different situations. While Glove Above focuses on recreational and emergency solutions for astronauts, Spock is arguably the 'cooler' of the two projects (and not just because of the Star Trek reference), not taking anything away from its usefulness and importance.

Spock incorporates voice control and gesture mapping, in order to deliver information to the astronaut, while in place. In the first prototype, which the team had come up with, in just over 24 hours mind you, they fitted a mobile phone along with some electrodes into a glove. The astronaut points to objects in space while wearing the glove ((named Spock) and asks about what he's pointing at. The glove also responds to other voice commands, the most important amongst which is of course the 'Take a selfie' command. There are less important commands that give astronauts information about various parts of their rovers etc. as well.

It's an innovative project to say the least and if the Amit Sharma and Arjit Sachdeva did all the work needed for it in about 24 hours,



Competitors present their projects in front of the judges.



Winners Amit Sharma and Arjit Saxena receiving their certificates from the judges.

one can only imagine what they would do with proper funding and time.

The same goes for Glove Above. The team took a regular woolen glove and attached Piezo Electric crystals to them, along with buttons on the fingers. On one glove, the buttons interact with an LED indicator, which displays messages like Help, Life etc. for astronauts to communicate in space. This is basically a prototype for what the group actually wants, which is a virtual keyboard. The other glove is for entertainment and plays various sounds in order to break the monotony in outer space. You can see both the space gloves on the International Space Apps Challenge website or on the videos on digit.in.

What's astonishing and at the same time commendable is that while both teams accomplished what they did over around a 24 hours period, they were actually allowed more time. The NASA Space Apps challenge is different from other hackathons in the sense that they

allow teams to start working on their projects even before the hackathon actually begins. Our winning teams chose the traditional hackathon way, but one wonders how much they would have done had they actually been working on this for a month or more.

You're probably thinking that none of this really makes a difference in your life right now. Well, the people we met there would disagree. Many of these youngsters will soon have their startups, they have a vision, something many of us spend entire lifetimes without. In addition, they have something over all of us, the talent to actually do what they're doing. And apart from the winners, we met a whole host of other talented young Indians filled with great ideas and willing to do something about it. We wish all of them lots of luck!

Winners Video: <http://digit.in/1FdjGpY>

Runners Up Video: <http://digit.in/1HjgIlz> ■